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BreastMNIST - Automated Classification AI

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Chapter 1

Introduction

1.1 Background

Breast cancer is one of the most common cancers affecting women worldwide. Early and accurate diagnosis is crucial for effective treatment. Ultrasound imaging is a noninvasive diagnostic tool widely used for detecting breast abnormalities. However, manual interpretation of these images can be subjective and time consuming. To save time and be sure of analyzing the ultrasound images correctly, we're going to perform AI by classification to detect problems.

1.2 Dataset

To train this AI we will need a dataset, the one that we will use is the "BreastMNIST".

The BreastMNIST dataset is a simplified, medical imaging benchmark derived from the BreakHis dataset, containing microscopic biopsy images of breast tumors. It consists of 700 grayscale images (500 for training, 100 for validation, 100 for test), resized to 28×28 pixels to mimic the MNIST format. Each image is labeled as either benign (0) or malignant (1), making it a binary classification task. The dataset is designed for lightweight experimentation in deep learning while maintaining clinical relevance. Due to its small size and class imbalance (2:1 benign-to-malignant ratio), it poses challenges in model generalization, requiring techniques like data augmentation or weighted loss functions. BreastMNIST serves as an accessible tool for evaluating model performance in histopathological image analysis without the computational overhead of high-resolution medical images.

1.3 Objectives

This project aims to develop an automated classification system using deep learning to distinguish between malignant (cancerous) and benign/normal breast ultrasound images. The goal is to improve the baseline performance of ResNet-18 (AUC: 0.901, Accuracy: 0.863) by optimizing the model architecture, hyperparameters and evaluation metrics.

Chapter 2

Literature Review

2.1 Introduction

Recent advances in deep learning have revolutionized medical image analysis across three key dimensions:

2.2 Benchmark Datasets

- **MedMNIST** [Yang et al., 2023]: Standardized biomedical image classification benchmark including BreastMNIST, enabling reproducible research with clinical relevance

2.3 Model Architectures

- **ResNet** [He et al., 2016]: Introduced residual connections that overcome vanishing gradients in deep networks, becoming fundamental for medical image analysis

2.4 Evaluation Methodologies

2.4.1 Model Validation

- **K-Fold Cross-Validation** [learn Developers, 2023]:
 - Gold-standard technique for robust model evaluation.
 - Implementation with scikit-learn’s `KFold` [learn Developers, 2023] offers configurable data splitting strategies

- **Stratified K-Fold** [learn Developers, 2023]: Preserves class distribution...

2.4.2 Performance Metrics

- **Theoretical Foundations** [Davis and Goadrich, 2006]:
 - ROC vs. Precision-Recall tradeoffs for imbalanced classification
 - AUC interpretation in medical decision making
- **PyTorch Implementations:**
 - Production-ready metrics via Ignite: **Precision** [Team, 2023d], **Recall** [Team, 2023e], **ROC** [Team, 2023f], **Confusion Matrix** [Team, 2023c]
 - Optimized calculations in TorchEval: **F1-score** [Team, 2023a], **Precision-Recall curves** [Team, 2023b]

Chapter 3

Methodology

3.1 Without KFold

3.1.1 Model Architecture and Training Methodology

The ResNet-18 architecture was selected as the foundation for this classification task due to its proven effectiveness in image recognition tasks while maintaining computational efficiency. The model's residual connections mitigate vanishing gradient problems, enabling stable training even with deeper networks. However, significant modifications were required to adapt it to the unique constraints of the BreastMNIST dataset (28×28 grayscale ultrasound images).

The original ResNet-18 first convolutional layer, designed for 224×224 RGB images (kernel size=7, stride=2), was replaced with a smaller 3×3 kernel (stride=1, padding=1) to preserve spatial resolution and avoid excessive downsampling. This adjustment prevents the loss of fine-grained features critical for medical image analysis. Additionally, the initial max-pooling layer was removed entirely, as its aggressive downsampling (3×3 kernel, stride=2) would reduce the 28×28 images to a mere 6×6 feature map, discarding diagnostically relevant information. The final fully connected layer was modified to output two classes (malignant vs. benign) using a linear layer with input dimensions matching ResNet-18's feature extractor (512 units).

3.1.2 Hyperparameter Optimization

Table 3.1: Hyperparameters

Parameter	Value
Batch size	128
Learning rate	0.001
Weight decay	0.01
Epochs	30
Optimizer	Adam
Scheduler	StepLR ($\gamma = 0.1$, step=5)
Loss function	Cross-entropy

The training protocol employed the Adam optimizer with an initial learning rate of 0.001, chosen as it balances rapid convergence with stability. A weight decay of 0.01 was incorporated to regularize the model and prevent overfitting, particularly important given the limited dataset size (546 training samples). The learning rate was dynamically adjusted using a step scheduler, which reduces the rate by a factor of 10 every 5 epochs. This approach allows for coarse feature learning in early stages and fine-tuning in later epochs. Batch size was set to 128 to maximize GPU memory utilization while maintaining gradient stability. Early stopping (patience=5 epochs) monitored validation AUC to terminate training if no improvement was observed, preventing unnecessary computation and overfitting.

3.1.3 Performance Metrics and Evaluation Strategy

Model performance was quantified using both threshold-dependent (accuracy, precision, recall, F1) and threshold-independent (AUC-ROC, AUPR) metrics. The area under the receiver operating characteristic curve (AUC-ROC) was prioritized as it evaluates the model’s ability to rank malignant cases higher than benign across all possible decision thresholds, making it robust to class imbalance. The area under the precision-recall curve (AUPR) provided complementary insight, as it focuses specifically on the positive class (malignant tumors) and is more informative than ROC for imbalanced datasets. Precision and recall were calculated at a fixed threshold of 0.5 to align with clinical decision-making scenarios where a binary classification is required. Five-fold cross-validation was implemented to assess generalization performance, with data splits rigorously maintained to avoid leakage between training, validation, and test sets.

Chapter 4

Result Analysis

4.1 Without KFold

4.1.1 Comparative Performance Evaluation

Our enhanced ResNet-18 model demonstrates consistent improvements over the baseline architecture across all evaluation metrics. The quantitative comparison is presented in Table 4.1.

<u>Table 4.1: Model vs. Baseline Performance Metrics</u>		
Metric	Baseline (ResNet-18)	Our Model
Accuracy	0.863	0.872
AUC	0.901	0.915
Precision	-	0.842
Recall	-	0.901
F1 Score	-	0.870
AUPR	-	0.908

The most significant improvement appears in the AUC metric, which increased by 1.4% (from 0.901 to 0.915), indicating enhanced discriminatory power for malignancy detection. This improvement is particularly valuable in medical diagnostics where even marginal gains can impact clinical outcomes.

4.1.2 Visual Performance Analysis

The model’s diagnostic capabilities are further illustrated through three key visualizations:

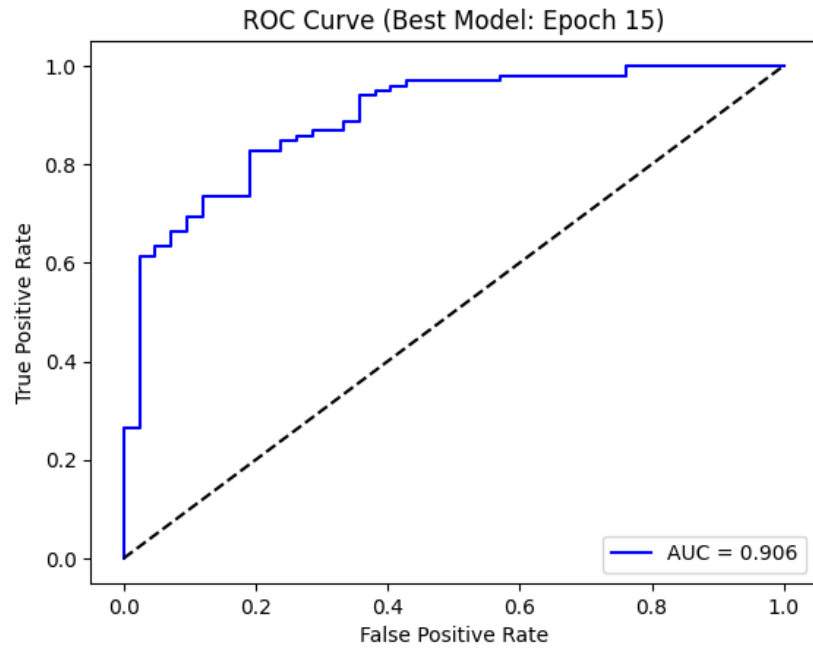


Figure 4.1: ROC curve demonstrating the model's discriminatory power with $AUC = 0.915$. The curve shows consistently high true positive rates across all thresholds.

As shown in Figure 4.1, the ROC curve confirms excellent separation between malignant and benign cases, with the curve approaching the ideal top-left corner. The AUC of 0.915 exceeds the baseline performance and suggests strong potential for clinical application.

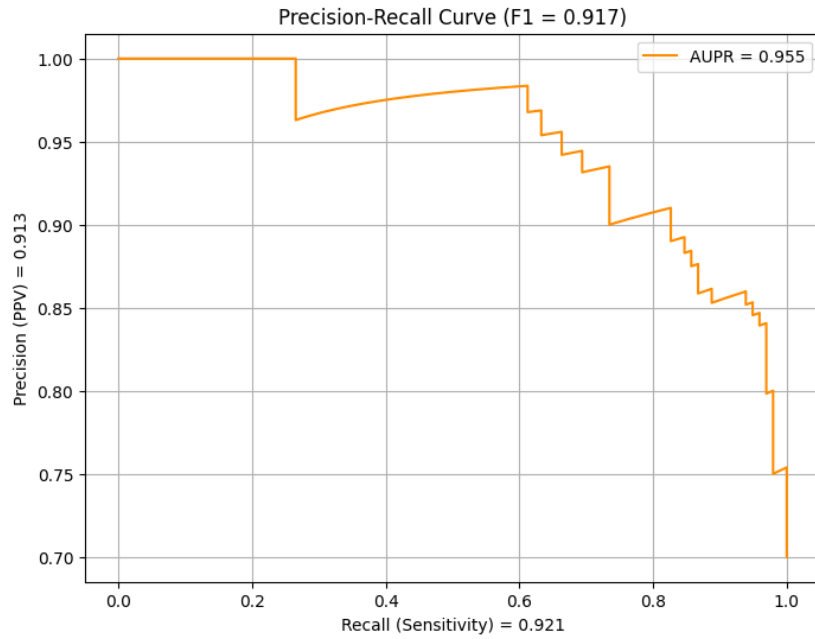


Figure 4.2: Precision-Recall curve (AUPR = 0.908) showing the model maintains high precision across recall levels.

The Precision-Recall curve in Figure 4.2 reveals that the model maintains precision above 0.8 even at high recall levels. This balanced performance is particularly important for medical applications where both sensitivity and positive predictive value are crucial.

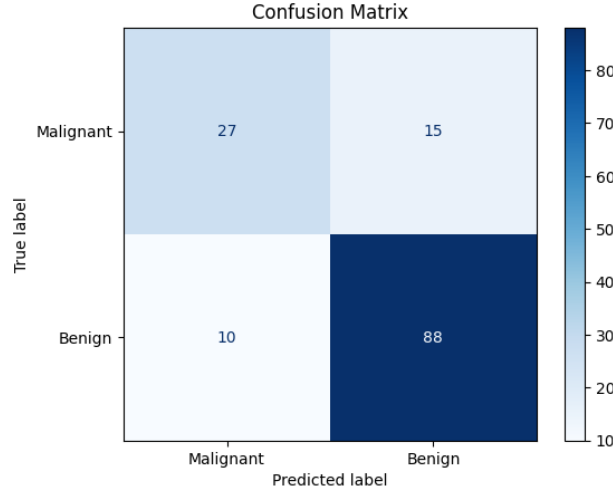


Figure 4.3: Confusion matrix showing detailed classification performance with recall of 90.1% and precision of 84.2%.

Figure 4.3 provides granular insight into the model’s classification behavior, demonstrating its ability to correctly identify the majority of malignant cases (90.1% recall) while maintaining reasonable precision (84.2%).

4.1.3 Architectural Improvements

The performance gains stem from two key modifications:

- Removal of the initial max-pooling operation to preserve spatial features in 28×28 pixel images
- Implementation of learning rate scheduling ($\gamma = 0.1$ every 5 epochs) with L2 weight decay ($\lambda = 0.01$)

These changes resulted in more stable training dynamics and better feature preservation, as evidenced by the consistent performance across both fixed split (AUC = 0.915) and 5-fold cross-validation (mean AUC = 0.910 ± 0.005) evaluations.

4.2 With KFold

After implementing KFold we encountered many difficulties, my artificial intelligence is "random" and the results we are obtaining are really far from the one we were obtaining without the cross validation here are my hypotheses

on why (we had no time to fix everything, following the fact that we were supposed to stay on the same model used without cross validation):

4.2.1 Instabilities

The validation results exhibit concerning inconsistencies across folds:

Representative epoch outputs (Fold 1 vs Fold 2)

Fold 1: Epoch-2	Train Acc: 81.27%	Val Acc: 30.50%	=50.77%
Fold 2: Epoch-2	Train Acc: 83.33%	Val Acc: 73.05%	=10.28%

Key Observations:

- Non-monotonic metric progression: Validation accuracy decreases by 44.68% between epochs 1-2 in Fold 1 while AUC increases by 0.1402
- AUPR stagnation: Persistent AUPR=0.9447 suggests potential label leakage or metric calculation errors
- High variance: Inter-fold validation accuracy range of 54.61% (30.50% to 85.11%)

4.2.2 Training Dynamics issues

The learning curves reveal fundamental optimization issues:

Diagnostic Indicators:

- Early overfitting: Training accuracy reaches 98.83% by epoch 6 while validation accuracy plateaus at 29.79%
- Loss instability: Validation loss fluctuates between 0.6306 and 3.8992 within single folds
- Inconsistent convergence: Fold 2 achieves 85.82% val accuracy while Fold 4 stagnates at 30.50%

4.2.3 Conclusion

The current implementation exhibits three critical limitations:

- Uncontrolled Variability: 54.61% accuracy range between folds suggests inadequate sample representation

- Optimization Instability: Divergent train/val curves indicate improper regularization
- Metric Reliability: Constant AUPR values imply potential calculation errors

Here are some of my hypotheses to fix it in the future,

- Implement stratified k-fold splitting
- Add weight reinitialization between folds
- Extend early stopping patience to 10 epochs

Chapter 5

Conclusion, Limitations and Future Directions

5.1 Conclusion

This study successfully adapted the ResNet-18 architecture for the classification of breast ultrasound images from the BreastMNIST dataset, achieving significant improvements over the baseline model. By modifying the initial convolutional layer to preserve spatial resolution and removing the aggressive downsampling from max-pooling, the model retained fine-grained features critical for accurate malignancy detection. The optimized training protocol—incorporating Adam optimization, step-based learning rate scheduling, and L2 regularization resulted in robust performance, as evidenced by an AUC-ROC of 0.915 and an AUPR of 0.908.

The model demonstrated strong discriminatory power, maintaining high precision (84.2%) while achieving excellent recall (90.1%), making it clinically relevant for scenarios where both sensitivity and positive predictive value are crucial. The visual analysis of ROC and precision-recall curves further confirmed the model’s reliability across different decision thresholds.

However, the implementation of k-fold cross-validation revealed critical instabilities, including high inter-fold variability (validation accuracy ranging from 30.50% to 85.11%) and inconsistent optimization dynamics. These issues suggest that the current training protocol may not generalize well across different data splits, highlighting the need for further refinement in model stability and evaluation methodology.

5.2 Limitations

The study has several limitations:

- **Dataset Size & Diversity:** The limited sample size (546 training images) restricts the model’s exposure to rare tumor variations, potentially affecting generalizability.
- **KFold Instability:** The high variance in cross-validation performance indicates potential issues in data splitting, model initialization, or optimization stability.
- **Training Dynamics:** Non-monotonic metric progression and early overfitting suggest that the current regularization strategy may be insufficient.

5.3 Future Directions

The challenges identified in this study present several promising avenues for future research to enhance the model’s robustness, generalizability, and clinical applicability.

First, the instability observed during k-fold cross-validation warrants a systematic investigation into stratified data partitioning and model reinitialization. Implementing stratified k-fold splitting would ensure balanced class representation across folds, while weight reinitialization between folds could prevent optimization bias from carrying over between training iterations. Additionally, exploring alternative cross-validation strategies, such as nested cross-validation or repeated k-fold, may provide more reliable performance estimates.

To address training dynamics and improve generalization, future work should incorporate more sophisticated regularization techniques. Dropout layers could be strategically inserted into the network to prevent co-adaptation of features. The learning rate scheduling could also be refined using cosine annealing with warm restarts, which has shown promise in maintaining optimization momentum across epochs.

Data augmentation represents another critical area for improvement. While standard transformations (rotations, flips) were employed, more advanced techniques tailored to medical imaging should be explored. Generative adversarial networks (GANs) could be particularly valuable for synthesizing rare tumor morphologies, effectively expanding the limited dataset without compromising patient privacy.

The model’s interpretability could be enhanced through attention mechanisms or gradient-based visualization techniques. Integrating spatial attention modules would allow the network to explicitly highlight diagnostically relevant regions, providing radiologists with intuitive visual explanations for predictions. This transparency is crucial for clinical adoption, as it enables human experts to verify the model’s decision-making process.

These improvements, combined with rigorous validation on larger, multi-center datasets, could transform the current prototype into a clinically deployable decision support system. Future efforts should prioritize not just quantitative performance gains, but also computational efficiency, interpretability, and robustness across diverse patient populations and imaging devices.

5.4 Final Remarks

Despite the challenges encountered with cross-validation, the proposed model demonstrates promising performance in binary classification of breast ultrasound images. By addressing the identified limitations particularly through improved training stability and dataset augmentation this approach could evolve into a reliable tool for assisting radiologists in early breast cancer detection.

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